

Part 1: Backend fundamental

1. What is Backend?

Backend is the server-side part of an application. It handles business logic, authentication, database operations, and sends responses to the frontend.

2. What is frontend?

Frontend is the user interface of an application. It is built using HTML, CSS and JavaScript and allows to interact with the application.

3

Frontend

User Interface
Runs in Browser

HTML, CSS, JS

Backend

Business Logic
Runs on server

Node.js, Java, Python

4. What is Client and Server?

Client: A client is a browser, mobile app, or device that sends requests to a server and displays the server's response.

Server: A server is a computer or software that receives client requests, processes them and sends responses back.

5 How do client and Server Communicate?

A client sends an HTTP request to the server. The server processes it, accesses the database if needed, and returns an HTTP response.

6 What is an API?

An API (Application Programming Interface) allows different software application to communicate and exchange data.

7 What is REST API?

A REST API follows REST principles and uses HTTP methods like GET, POST, PUT, PATCH, and DELETE to perform CRUD operation.

8 What is HTTP?

HTTP (HyperText Transfer Protocol) is the protocol used for communication between clients and server.

9 What is HTTPS?

HTTPS is the secure version of HTTP. It encrypts data using SSL/TLS to protect information during transmission.

10. What is Request and Response?

Request: A request is a message sent by the client to the server asking for data or action.

Response: A response is the data or result sent by the server back to the client after processing a request.

11. What is CRUD?

CRUD stands for Create, Read, Update, and Delete, which are the four basic database operation.

12. What is JSON?

JSON (JavaScript Object Notation) is a lightweight format used to exchange data between the frontend and backend.

13. What is XML?

XML (Extensible Markup Language) is a markup language used to store and transfer data, but today JSON is more commonly used in web application.

13. What is Database and types?

A database is a system used to store, organize, manage, and retrieve data efficiently.

There are two ^{main} types of database.

- SQL database (MySQL, PostgreSQL)
- NoSQL database (MongoDB, Cassandra).

14. SQL vs NoSQL

SQL	NoSQL
Tables	Documents
Fixed Schema	Flexible Schema
MySQL	MongoDB

25 What happens when you type google.com

The browser finds Google's IP address using DNS, sends an HTTP request to the server, the server processes it and returns HTML, CSS, and JavaScript, which the browser renders as the webpage.